

Als or Humans in Luxury Shopping? A Design Study of VR Shopping Assistants

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ABSTRACT

This study investigates four shopping assistant types (3D avatar, webcam, AI, voice) in VR luxury environments through user studies with 20 participants using NASA-TLX, questionnaires, and interviews. Results show 3D avatar assistants excelled in immersion, trust, and satisfaction, while human-operated assistants outperformed AI in realism and satisfaction. Webcam achieved 50% immersion preference versus 75% for 3D avatars despite strong trustworthiness. The study pioneers PC-VR asymmetric interaction systems, providing design insights and optimization strategies for luxury retail VR shopping assistants.

Index Terms: Virtual Shopping, Avatars, Cross-device user interaction, Social Virtual Reality, Shopper-Buyer Communication.

1 INTRODUCTION

Virtual reality (VR) technology offers unprecedented opportunities for luxury retail by enabling immersive 3D product visualization and personalized service experiences [3]. However, the design of shopping assistants for replicating the consultation and expertise of human sales associates still remains underexplored. Existing research primarily compares individual assistant types [2] but lacks systematic evaluation across multiple modalities within the same luxury shopping context.

This study addresses three research questions: (1) How do different assistant types (3D avatar, webcam, AI, voice) influence consumer experience in VR luxury shopping? (2) How do human-operated versus AI assistants differ in trust and satisfaction? (3) What design factors affect user acceptance of webcam-based asymmetric PC-VR interaction?

We implemented a networked VR luxury store with four assistant conditions and conducted a within-subjects study (N=20) using NASA-TLX, custom questionnaires, preference rankings, and semi-structured interviews. Our findings reveal that 3D avatars provide optimal user experience, while webcam assistants demonstrate a fundamental tension between realism and immersion in VR environments.

2 RELATED WORK

Prior work shows that photorealistic 3D avatars enhance presence and reduce cognitive load in virtual retail. Voice assistants can increase attention but may not improve decision-making. AI-driven chatbots offer personalized recommendations but face challenges in emotional connection [4]. However, comprehensive comparisons across assistant types in luxury contexts remain limited. In addition, recent studies explore webcam-animated avatars in videoconferencing, showing improved meeting satisfaction. Yet their effectiveness

as shopping assistants in immersive VR is unexplored, particularly regarding dimensional integration with 3D environments. Furthermore, research indicates human frontline employees outperform AI for experiential products requiring subjective judgment, while AI excels in efficiency-focused tasks [1].

3 SYSTEM DESIGN

We developed a multiplayer VR luxury shopping system using Unity 2022.3.12f1, Meta Quest 3 headsets, and Mirror networking framework. The environment features photorealistic luxury products (e.g., designer bags) in a high-end retail setting with interactive product manipulation (grabbing, rotating, scaling).

3.1 Four Assistant Types

We implemented four assistant types to systematically vary embodiment and control mechanisms. The 3D Avatar Assistant featured a rigged photorealistic female character operated by a human via VR controllers, enabling full mobility, gesturing, speech, and physical product manipulation. In contrast, the Webcam Assistant presented a real-time video feed from a PC webcam displayed on a 2D canvas within the VR environment, where the human assistant's face appeared on screen with authentic facial expressions but no manipulation capabilities. The AI Assistant (GPT-4) employed a 3D avatar driven by GPT-4 for speech-to-text and text-to-speech conversion, achieving an average response latency of 2.3s, with text overlays displaying responses and the avatar featuring lip-sync. Finally, the Voice Assistant provided audio-only communication via spatial voice chat, eliminating visual representation entirely and relying solely on auditory cues for interaction.

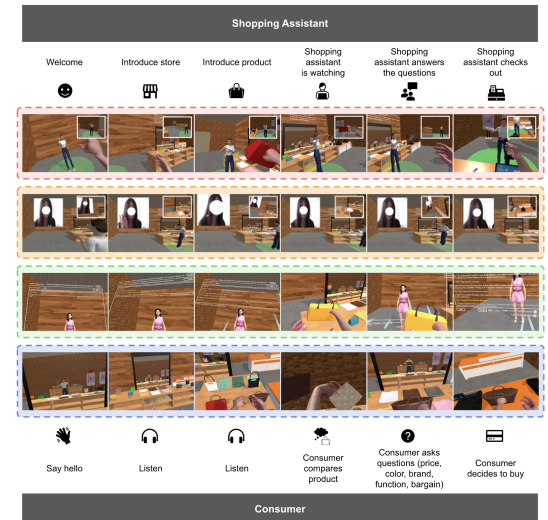


Figure 1: User study progress of four shopping assistants in VR: 3D avatar, webcam, AI, and voice

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3.2 STUDY DESIGN

We recruited 20 university students (10M/10F, aged 24-29) through campus channels. The within-subjects design used Latin Square counterbalancing across four conditions (10 min each) in a simulated luxury shopping scenario. After each condition, participants completed NASA-TLX and a custom 20-item questionnaire. After that, they completed preference rankings across 10 dimensions and semi-structured interviews.



Figure 2: User participating in the user study: (1) Demographic questionnaire; (2) Warm-ups and tutorials; (3) Task with different conditions (a.voice, b.webcam, c.3D avatar, d.AI assistant); (4) Post-task questionnaire; (5) Interview.

4 RESULTS

NASA-TLX revealed significant differences in Performance ($F(3,57)=2.80, p < .05$) and Frustration ($F(3,57)=2.92, p < .05$), with 3D avatars ($M=6.00$) outperforming AI ($M=4.75, p < .05$) while showing lower frustration ($M=1.90$ vs. $3.35, p < .05$). Across 20 evaluation dimensions, 3D avatars significantly surpassed AI in 11 metrics ($p < .05$ or $p < .01$), including Immersion, Attention Focus, Natural Realism, Service Trust, and Overall Satisfaction. Notably, 3D avatars dominated long-term usage intent ($M=5.35$) over webcam (3.95), AI (3.30), and voice (4.00), $p < .01$. Webcam assistants excelled in interaction fluency ($M=5.40$) and trust ($M=5.45$) but underperformed in immersion ($M=4.45$). Preference rankings showed 3D avatars leading with 55.5% average preference across 8/10 dimensions, including Immersion (75%), Presence (70%), and Satisfaction (60%), while voice assistants showed strength in cognitive load reduction (35%).

Thematic analysis revealed five key themes: (1) Immersion-Realism Trade-off, 14/20 participants experienced "dimensional misalignment" with webcam, perceiving it as "a window to the real world" rather than integrated VR (P14: "felt like a remote connection"); (2) Social Pressure, 16/20 became more polite with webcam versus casual with AI assistant and 3D avatar assistant; (3) Context Dependency, 17/20 preferred human assistants for luxury goods, 18/20 preferred AI for embarrassing products; (4) Trust Mechanisms, 9/20 trusted AI for objective information, 15/20 trusted humans for emotional support; (5) Technical Limitations, AI response delays (8/20) and webcam positioning inflexibility (12/20).

5 DISCUSSION

Our findings demonstrate that 3D human-controlled avatars provide an optimal balance of immersion, trust, and satisfaction in VR luxury shopping environments. Across all evaluation dimensions, human assistants (3D, webcam, voice) significantly outperformed AI in overall satisfaction ($M=5.42$ vs. $4.05, p < .01$), validating the critical importance of human agency for experiential products where emotional connection and personalized service are paramount. The 3D avatar condition particularly excelled in long-term usage intent and dominated preference rankings across eight of ten dimensions, suggesting that the combination of spatial presence and human intelligence creates a compelling service experience.

The webcam condition revealed a fundamental design tension in VR assistant implementation. While authentic facial expressions achieved high trustworthiness and service quality ratings, the integration of 2D video into a 3D virtual environment disrupted spatial immersion. This "immersion-realism trade-off" demonstrates how dimensional misalignment undermines presence despite enhanced authenticity. Participants described experiencing the webcam as "a window to the real world" rather than an integrated VR element, highlighting that visual realism alone cannot compensate for spatial incongruence in immersive environments.

Based on these findings, we recommend several practical guidelines for VR retail implementation. First, prioritize 3D avatars for luxury retail contexts requiring high immersion and emotional engagement. Secondly, use webcam assistants selectively for trust-critical consultations where authenticity outweighs spatial coherence, such as high-value purchase decisions. Third, improve AI responsiveness to achieve lower latency of communication and more interactive behavior with consumers. Finally, implement hybrid strategies that dynamically adapt assistant type to product category and shopping stage, leveraging each modality's strengths.

6 LIMITATION

Several limitations constrain the interpretation and generalizability of our results. First, conditions are inherently confounded as assistant types differ in manipulation capabilities and communication modalities, making it difficult to isolate individual factors. Next, our student sample ($N=20$) does not represent actual luxury consumers, limiting both statistical power and ecological validity. Third, the non-photorealistic virtual environment may underestimate the effects of visual realism on immersion and presence. Finally, the single-session design prevents assessment of long-term adaptation effects, learning curves, or potential shifts in preference with repeated exposure to different assistant types.

7 CONCLUSION

This exploratory study provides the first systematic comparison of four VR shopping assistant types in luxury retail contexts. Results demonstrate 3D human-controlled avatars as the optimal solution, while revealing fundamental trade-offs between realism and immersion in asymmetric PC-VR interactions. Future work should address confounds through orthogonal manipulation of embodiment and capabilities, recruit diverse luxury consumers, and develop adaptive hybrid systems combining assistant strengths across shopping phases.

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